AKSHATA NALAWADE

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LinkedIn | GitHub | Portfolio: akshata-portfolio-mu.vercel.app

PROFESSIONAL SUMMARY

Electronics and Computer Science engineering student building applied AI systems across generative models, computer vision, and conversational AI. Built House Muse, a full-stack AI interior design platform that redesigns a room from a single photo using SDXL with Depth ControlNet and style LoRAs on top of a YOLOv8, SAM, and CLIP vision pipeline, and shipped a lead-handling chatbot on Zoho SalesIQ during an AI chatbot development internship. Certified in prompt engineering; comfortable with Python, PyTorch, and serving ML models behind REST APIs.

TECHNICAL SKILLS

Generative AI: Stable Diffusion XL, ControlNet (Depth), LoRA, img2img, Prompt Engineering
Computer Vision: YOLOv8, SAM, CLIP, OpenCV, MediaPipe, Grad-CAM
Machine Learning: PyTorch, CNN, Transfer Learning (MobileNetV2, ResNet18), KNN, K-Means, Pandas, NumPy
Conversational AI: Zoho SalesIQ, Intent Mapping, Decision-Tree Conversation Flows, Workflow Automation
Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3
Web & APIs: React, Vite, Tailwind CSS, Node.js, Django, Django REST Framework, Flask, RESTful APIs, Token Auth, Google OAuth
Data, Cloud & Tools: PostgreSQL, SQLite, Hugging Face Spaces, Linux, Cloud Computing, Git, GitHub, Jupyter, Colab, Figma

EDUCATION

Vidyalankar Institute of Technology *Aug 2023 – Aug 2027*
B.Tech in Electronics and Computer Science | Avg. GPA till Sem V: 7.61
Guru Nanak Khalsa College of Arts, Science & Commerce | Maharashtra State Board (HSC) 75.50%
IES V. N. Sule Guruji English Medium High School | Maharashtra State Board (SSC) 72.40%

EXPERIENCE

AI Chatbot Development Intern – Trinetrini Quantum *1 Dec – 15 Jan*

- Built a conversion-focused chatbot on Zoho SalesIQ that automates website lead handling.
- Mapped user intents and designed decision-tree conversation flows to guide visitors through common queries.
- Owned the Zoho integration, workflow automation, testing, and deployment.
- Structured user interaction handling to cut repetitive manual responses and improve automation efficiency.

Tech: *Zoho SalesIQ, Conversational AI, Workflow Automation, Decision-Tree Logic, JavaScript*

KEY PROJECTS

House Muse – AI Interior Design Platform *GitHub*

- Built a full-stack platform that turns a room photo into a photorealistic redesign while keeping walls, windows, and layout intact, using SDXL with Depth ControlNet and style LoRAs.
- Implemented a vision pipeline (YOLOv8, SAM, CLIP, K-Means) for object detection, room-type classification, lighting analysis, and colour palette extraction.
- Designed a three-service architecture: React + TypeScript frontend, Django REST backend with token auth and Google OAuth, and a Flask ML microservice.
- Added chat-driven image refinement using img2img, and catalog-based budget breakdowns in ₹ across 7 categories with an interactive donut chart.

Tech: *PyTorch, SDXL, ControlNet, LoRA, YOLOv8, SAM, CLIP, Hugging Face, Flask, Django REST Framework, React, TypeScript*

Weather Classification – Deep Learning Image Classifier *GitHub*

- Built and compared three CNN classifiers (MobileNetV2, ResNet18, and a custom CNN) for classifying weather from sky images, evaluating each with confusion matrices to select the best model.
- Added Grad-CAM explanations to show which image regions drive each prediction, and served the model through a Flask app.

Tech: *Python, PyTorch, CNN, Transfer Learning, Grad-CAM, Flask*

IshaaraSpeak – Audio-to-Sign Language System

- Designed a system that converts speech to text and generates 3D sign language output, aimed at accessibility.
- Worked on real-time audio processing and gesture mapping using CNN, OpenCV, and MediaPipe.

Tech: *Python, JavaScript, CNN, OpenCV, MediaPipe, 3D Animation*

Other projects: Creamy Bliss (full-stack web app for an ice cream parlour; Node.js, PostgreSQL) · NetLab (browser-based simulator for six data-link protocol experiments; JavaScript)

CERTIFICATIONS & TRAINING

Introduction to Prompt Engineering | Certificate code: 9515491 *2 Dec 2025*
Prompt Engineering Applications | Certificate code: 9515955 *2 Dec 2025*
Fundamentals of Machine Learning – Infosys Springboard *Completed*
Certificate Course in Cloud Computing – C-DAC ACTS, Pune (60 hours: cloud, Linux, deployment) *5 June – 15 June*